



COURSE DESCRIPTION		
Course Code	26SE104	Course Credit: 4
Course Name	Advanced Mathematical Methods for Computing Systems	
Course Delivery Mode	TiC & PiL	
L-T-P-H	3-0-2-0	
Pre - requisite	NIL	
Year of Introduction and Syllabus Version	2026 V1.0	

COURSE OBJECTIVES		Cognitive Level	Knowledge Category
1	To develop a strong foundation in advanced linear algebra techniques for analysing computing systems	L3	K3
2	To understand and apply optimization methods to solve engineering and machine learning problems	L4	K4
3	To gain knowledge of probability theory and statistical methods for modelling uncertainty in real - world applications	L3	K3
4	To integrate mathematical concepts with computational tools to solve complex problems in data science and engineering	L4	K4

COURSE OUTCOMES (COs)	
After the successful completion of the course students will be able to:	
CO1	Apply advanced linear algebra concepts such as eigenvalues, orthogonality, and matrix decompositions in computing applications
CO2	Formulate and solve optimization problems using appropriate analytical and numerical techniques
CO3	Model and analyse random phenomena using probability distributions and statistical inference methods.
CO4	Implement mathematical methods using computational tools to solve real - world engineering and data - driven problems

CO - PO MAPPING						
	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3	-	2	1	2	1
CO2	3	-	2	1	3	1
CO3	2	-	2	1	2	1
CO4	2	-	2	1	1	1
Correlation levels: 1 - Low; 2 - Medium; 3 - High; No Correlation - " - "						



COURSE IMPLEMENTATION CLOCK (IN HOURS)	
Zeroth Module	1 Hr.
Course content delivery hours	60 Hrs.
Internal Test (2 Assessment Tests of one and a half hour duration)	3 Hrs.

COURSE CONTENTS

Zeroth Module: General Introduction to the Course		
- Course introduction with a thorough briefing of Course Objectives and Course Outcomes - An overview of the POs that are mapped to the course	No ESE	1 Hr.

Module - I		COs	Hrs.
Module Title	Advanced Linear Algebra for Computing		
Vector spaces - Subspaces - Basis - Dimension - Linear transformations and matrix representations - Eigen values - Eigenvectors - Diagonalization - Singular value decomposition (SVD) - LU decomposition - QR decomposition - Iterative methods (Jacobi, Gauss-Seidel) - Least squares problems		1,4	12
Module - II		Cos	Hrs.
Module Title	Optimization Techniques		
Formulation of optimization problem - Convex sets and convex functions - Unconstrained optimization - Gradient descent and variants - Newton's method - Constrained optimization - Method of Lagrange multipliers - Kuhn-Tucker (KKT) conditions		2,4	12
Module - III		Cos	Hrs.
Module Title	Probability Theory and Statistics		
Random variables (discrete and continuous) - Probability distributions (normal, binomial, Poisson, exponential) - Expectation - Variance - Joint distributions and independence - Sampling methods - Parameter estimation (MLE, MAP) - Hypothesis testing - Confidence intervals - Regression analysis (linear)		3,4	12

List of Exercises (Minimum 8 exercises to be completed)		CO
1	Introducing Python as a tool for computational mathematics.	1,2,3,4
2	Introducing different libraries of Python, which serves as effective tools in solving problems.	1,2,3,4
3	Performing vector and matrix operations using Python and visualizing the results	1,4
4	Solving systems of linear equations using matrix methods and visualizing the solutions.	1,4
5	Computing eigenvalues, eigenvectors and Singular Value Decomposition (SVD) and visualizing their applications.	1,4
6	Implementing the gradient descent algorithm to solve optimisation problems and visualise convergence.	2,4

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7	Solving optimization problems using Newton's method and visualizing results.	2,4
8	Solving constrained optimization problems using Lagrange multipliers and visualizing solutions.	2,4
9	Simulating random variables and probability distributions and visualising their outcomes.	3,4
10	Performing hypothesis testing and regression analysis using Python and visualizing the results.	3,4

LABORATORY MANUAL

An internally prepared laboratory manual contains

- Vision and Mission of the institution
- Details of laboratory equipment
- Details of laboratory exercises

TEXT BOOKS

1	G. Strang, <i>Linear Algebra and Its Applications</i> , 4 th Edition, Cengage Learning, 2006, ISBN: 9780030105678
2	S. Boyd and L. Vandenberghe, <i>Convex Optimization</i> , 1 st Edition, Cambridge University Press, 2004, ISBN: 9780521833783.
3	S. Ross, <i>A First Course in Probability</i> , 9 th Edition, Pearson Education, 2014, ISBN: 9780321794772

REFERENCE BOOKS / WEB REFERENCES

1	D.C. Lay, S.R. Lay, and J.J. McDonald, <i>Linear Algebra and Its Applications</i> , 5 th Edition, Pearson Education, 2015, ISBN: 9780321982384.
2	K. Hoffman and R. Kunze, <i>Linear Algebra</i> , 2 nd Edition, Prentice Hall of India, 2006, ISBN: 9788120302709.
3	.S. Rao, <i>Engineering Optimization: Theory and Practice</i> , 4 th Edition, Wiley, 2009, ISBN: 9780470183526.
4	D.P. Bertsekas, <i>Nonlinear Programming</i> , 2 nd Edition, Athena Scientific, 1999, ISBN: 9781886529007.
5	W. Feller, <i>An Introduction to Probability Theory and Its Applications</i> , Vol. 1, 3 rd Edition, Wiley, 1968, ISBN: 9788126518050.

MODES OF EVALUATION	SCORE WEIGHTAGE / SPLIT MARKS
Continuous Internal Evaluation (CIE)	50
Internal Test - I	7.5
Internal Test - II	7.5
Taskworks (Assignments/MCQ/Quiz)	10
Embedded Lab	25
End Semester Examination (ESE)	50
Total	100
End Semester Examination Pattern	
The question paper consists of two parts: Part A: 14 marks Part B: 36 marks Part A contains a total of 7 questions, 2 questions each from 2 modules and 3 questions from 3 rd module, having two marks for each question. Part B contains 2 questions from each module of which student should answer any one. Each question can have a maximum 2 subdivisions and carry 12 marks	





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SAINTGITS COLLEGE OF ENGINEERING (AUTONOMOUS)
M.Tech. SYLLABUS 2026

APPROVAL SUMMARY	
	Number and Date
Recommended by BoS	SCE/BoS/CS/M.Tech./SE/013/25- 26, 22-05-26
Approved by Academic Council	AC/RES/019/25-26, 15-06-2026

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